Towards Ethical Moral Machines

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***Abstract*— This work extends recent state of the art in modeling human moral decision making by using mathematical and philosophical theory to suggest adaptations to existing models. This research analyses alternatives to the normal distribution when attempting to model ethical principles. We show how the ethical theories of utilitarianism and deontology can be combined into an informative prior distribution. We continue to expand state of the art models to consider ethical dilemmas beyond the Trolley Problem and show how the models must be adapted to address complex scenarios. This research also considers how models of morality could be aggregated to automate ethical decision making and question how representative these voting systems can be. These adaptations are not necessarily intended to improve the model but are aimed to raise awareness about the importance of interpreting the results relative to the assumptions either implicitly or explicitly made in the model.**

I. INTRODUCTION

Modern artificial intelligence (AI) is being applied to contexts that have dramatic social implications. We are seeing the introduction of AI in potential life or death situations, the criminal judiciary system and positions of care for those most vulnerable in society. As computers become increasingly prevalent, the subject of designing intelligent systems that function responsibly is increasingly important [1]. A recent report published by the House of Lords Select Committee on Artificial Intelligence on 16th April 2018 [2] recognises the importance of ethical consideration and recommends the focus of funding in this area.

The report highlights autonomous vehicles (AV) as an area requiring urgent consideration, focusing on how AV make ethically sensitive decisions. The recent fatal accident in Arizona involving Uber's unmanned vehicle and a pedestrian fueled doubt over the ethicality these vehicles. In 2017, the UK Government pledged a significant proportion of its 70 million AI budget to having fully automated cars in use by 2021. However, earlier in 2018, the UK government announced a 3 year review to assess the risks of AV before they are tested on British roads.

The challenge of designing intelligent systems that are capable of making semantically interpretable decisions is captured by the argument over three types of knowledge: explicit, implicit and tacit. Explicit knowledge is the type of knowledge that can be extracted from an individual over suitable enquiry [3]. Implicit knowledge, on the other hand, is made up of "what we know but aren't aware of" [4]. Tacit knowledge refers to knowledge that is unable to be articulated and "tied to the senses" [5]. Ethics is a form of tacit knowledge. We are unable to explain why we think, or act, a certain way other than an intuition that guides our decisions over ethical dilemmas. Much literature has been dedicated to the problem of extracting and understanding tacit knowledge from human behaviour [5] [6]. However, creating definitions over tacit knowledge is extremely difficult: how do we explain what we do not understand? This seemingly unsolvable paradox significantly affects the field of engineering ethical machines, how are we to bridge the semantic gap between human ethics and codifying ethical principles?

Wallace and Allen summarise approaches to automating ethics as either top-down or bottom-up [7]. A top-down approach refers to the process of iteratively reducing a problem into individually solvable sub tasks. Charisi et al. [8] identify the most frequent form of top-down approach being used to implement ethical behaviour as system governance via a set of ethical rules. This approach is adopted by Arkin et al.'s "ethical governor" [9], a component of the ethical architecture for a military autonomous system. The ethical governor's purpose is to conduct an evaluation of the "ethical appropriateness" [9] of an action prior to its completion by a robot. The governor ensures a response is ethical by ensuring its non-membership of a pre-defined set of possible unethical outcomes. The "ethical appropriateness" of a response is determined by both utilitarian and deontological ethical theory. The critical debate between ethical theorists complicates top-down approaches to autonomous ethical decisions and has been interpreted differently throughout literature [[10], [11]]. Top-down approaches like these allow for a rigorous process of decision making [8]. However, ambiguity over ethical theory means that it is not clear as to whether top-down approaches dependent on specified ethical theory can be used in practice. Attention has, therefore, turned towards bottom-up approaches that do not require hard coded ethical rules [8]. Bottom-up approaches in engineering require a description of a problem and then the development of a method to find a solution in terms of its parameters [8]. In their general ethical analyzer, "GenEth", Anderson and Anderson use machine learning to learn an ethical theory [12]. GenEth was created alongside input from ethicists to help codify ethical principles in any given domain. In order to learn ethical principals GenEth uses inductive logic programming, a technique that learns relations based on First Order Horn Clauses. Contrastingly, Abel et al. use reinforcement learning to learn the most moral decision. Abel et al. use Markov Decision Processes and Partially

Observable Markov Decision Processes as mechanisms to frame a decision problem with an associated reward function. Contrastingly, D.Dewey in his work, "Learning What To Value" [13], argues that reinforcement learning can only learn preferences based on potential rewards. Dewey continues to present "expected observation utility maximization" as a mechanism that, unlike reinforcement learning , can be used to define agents with multiple final goals. Similarly, Bostrom in his recent work, "contemplating the feasibility of super-intelligence" [14], also questions the use of reinforcement learning in learning moral theory. Bostrom suggests that a sufficiently intelligent machine could maximise its reward by exploiting or "wireheading" [14] its reward function. Although only applicable to some reinforcement learning scenarios, this vulnerability shifts attention to Bayesian approaches.

Both Bostrom [14] and Dewey [13] suggest utility functions as a preferential way of ensuring that an AI learns about moral values instead of decision outcomes. Bayesian learning is an alternative bottom-up technique that allows agents to make decisions that optimize a meta-utility function [15]. In their paper, "Learning a Common Sense Moral Theory" [16], Kleiman-Weiner et al. introduce a novel computational framework for understanding and learning moral theory. They first introduce a recursive utility calculus that captures welfare trade-offs in interactions between individuals. They then use hierarchical Bayesian inference as a mechanism to understand the moral actions and decisions of individuals. Kleiman-Weiner et al. define abstract principles that capture simplified relationships between individuals and explain why a particular individual may act towards another. They propose a structured model where each individual's principles are generated from a prior, dependent on the group that the individual belongs to [16]. Building on this theory, the paper, "A Computational Model of Commonsense Moral Decision Making" [17] introduces a novel computational model of moral decision making in the AV domain. Kim et al.'s model describes moral dilemmas as utility functions that use abstract moral principles to compute trade-offs in decision making. Kim et al. use a Bayesian hierarchical model to categorise social structures of individuals and groups and to show that individual moral preferences can be inferred as interpretable parameters from limited data.

This work is focused on the challenge of automating ethical decision making. Some critics argue that machines will never be moral beings [18], yet systems that have to make ethical decisions are being used today. We proceed by analysing state of the art in this area that use models of human moral behaviour to help define ethical behaviour of autonomous systems [[17], [16]]. We extend these works by first presenting potential pitfalls of their approaches, accompanying these hypotheses with experimental results. The focus of this project therefore, is not to propose a solution to ethical automation but to highlight issues and raise questions that must be addressed in order to drive future progress in this field.

We begin by examining the Moral Machine dataset, employed by Kim et al. in their model, the first of its kind to crowd-source morality on a large scale [19] and question the ethicality of gamifiying this form of data collection. We present an alternative moral behaviour dataset, collected by a German autonomous vehicle study . [20], used in this work's model implementation.

We continue to adapt the method of Kim et al. to build a model of moral decision making around the Faulhaber dataset, evaluating the model using Monte Carlo heuristics. Our results show that our implemented hierarchical model achieves 82% predictive accuracy over our dataset. However, this paper's emphasis is not on the quantified success of the model but on the questions and issues our implementation has raised. We therefore challenge assumptions made by Kim et al. and highlight how these affect results. Continuing our line of questioning, we consider utility calculus as a mechanism to capture human morality. We explore the philosophical debate surrounding the use of moral principles in decision making and demonstrate how experimental results could be used to ground ethical theory. Further, we discuss the link between deontology and utilitarianism, showing, through our implementation, how deontological statements can be used as a prior for the utilitarian model to achieve greater predictive accuracy. We continue to test the model of Kim et al. on moral dilemmas that extend the trolley problem to consider more complex cases of morality and find that the future of ethical machines is heavily dependent on mathematics. Returning our attention to automating ethical decision making, we discuss the feasibility of using a model of human moral behaviour within these intelligent systems. We conclude by using the theory of Noothigattu et al. to assess the potential of using an aggregation of a model of morality to automate ethical decision making.

II. MIT Moral Machine Dataset

The majority of historical work on ethical machines is concerned with building theoretic models that formulate moral theory. However, if we are to succeed in building machines capable of making ethical decisions then these models need to be tested in real world contexts. There is an urgent need for data that encapsulates information about how humans make moral decisions. Francis et al. [21] identify moral dilemmas as key in helping researchers understand moral decision making and use their study to collect data about human decision making in the most famous of all moral dilemma thought experiments, the trolley problem. Francis et al. use virtual reality as an immersive technique to compare participants' responses to a variety of trolley problem scenarios. Contrastingly, researchers at MIT take a different approach to collecting

trolley problem data. Their platform, The Moral Machine, is the first of its kind to "crowdsource morality" [19]. As of October 2017 the platform had collected over 30 million responses from over three million respondents from over 180 countries from across the world [17].

The Moral Machine dataset has motivated further research [17], [22], who have used the data to learn underlying moral preferences. However, recent studies have warned of the dangers of using unqualified big data to hype experimental results [23]. As attention turns towards implementing real systems that make ethical decisions we need to ask the right questions concerning the data that these systems may be trained on. One particular concern with the data generated by the MIT Moral Machine platform is the gamification of data collection. Dergousoff et al. raise questions surrounding the quality of results collected through a gamified approach to data collection when compared to traditional methods [24]. Similarly, Versteeg establishes gamification as a potentially "manipulative construct" [25] which reinforces our own opinions about the Moral Machine's data quality: are participants taking it seriously? What happens if someone gets bored in the middle of playing? Do people understand that this data is being used in academic research? We opt to select a new dataset which allows us to question the repeatability of Kim et al.'s results [17]. Additionally, we select a dataset that has been collected in a traditional laboratory environment to avoid the ethical problems we have raised over the Moral Machine dataset.

III. Autonomous Vehicle Dataset: Wachter et al.

The data used for this project's implementation is from the German study "Human Decisions in Moral Dilemmas are Largely Described by Utilitarianism: Virtual Car Driving Study Provides Guidelines for Autonomous Driving Vehicles" [20] and is referred to as the "German Autonomous Vehicle" dataset for the remainder of the article. Faulhaber et al. conduct a set of experiments in which participants experience modified trolley problems as the driver of a car in a virtual reality environment. Participants are forced to make one of two decisions, choosing between the left or right lane when faced with obstacles. The obstacles consist of a variety of human-like avatars of different ages and numbers. Each participant is presented with one training track and five different experiment tracks. The car being driven by the participant is traveling at 36 mph and the tracks range in length between 180m to 200m. The experiment tracks consist of five different environments: two mountain, a suburban and two city levels. In the city scenarios, the participant has to decide whether to swerve and mount the pavement or continue on the road. Example scenarios are shown in Figure 1. The presented avatars are: middle aged man, old man, young boy, kneeling man, self (the participant who is driving). The avatars are all male, range in number and presented in random combinations. Prior to starting the experiment Faulhaber et al. ensure each participant is aware of the nature of the experiment and make sure they have

Fig. 1. Example scenario from the German Autonomous Vehicle Study [20]

signed a consent form clarifying they are able to terminate the experiment at any time. Each participant is then presented with a sequence of training scenarios followed by 24 test scenarios across a combination of the five different test environments. Finally, the participants are asked a series of questions to ascertain age, gender and driving experience alongside a questionnaire containing high level philosophical questions. 216 unpaid participants take part in the study and 201 participants complete the study.

IV. Methodology

To evaluate the success of Kim et al.'s model on the German Autonomous Vehicle Dataset we follow the method of Kim et al. [17]. However, for our implementation, the mapping between observable objects onto the abstract feature space is altered slightly due to the differences between the Moral Machine dataset and The German Autonomous Vehicle dataset. The characters represented by the German Autonomous Vehicle dataset are: driver of the vehicle (the respondent), man, old-man, young boy, kneeling man and pedestrians. Our character vector is therefore represented as $\Theta_y \in \mathbb{N}^K$ where $|K| = 6$. Following Kim et al.'s approach, we believe that each of the six characters can be described by one or more of the following abstract principles: self-preservation, elderly, infancy, middle-aged, kneeling-down, pedestrian. Our implementation therefore uses the linear mapping $F(\Theta) = A'\Theta$ where A' is a 6x6 matrix.

V. Composing Utility Functions

Kim et al. choose to model the utility value of a resultant state as a linear combination of the features in the abstract dimension,

$$\mu(\Theta_i) = w^\top F(\Theta_i), \tag{1}$$

where Θ_0 represents the state achieved from choosing not to switch lanes and Θ_1 represents the state achieved by choosing to switch lanes. A respondent's decision to switch lanes, $Y = 1$, is represented by the sigmoid function of net utility of the two choices [17],

$$P(Y = 1|\Theta) = \frac{1}{1 + \exp^{-U(\Theta)}}, \tag{2}$$

$$U(\Theta) = \mu(\Theta_1) - \mu(\Theta_0). \tag{3}$$

VI. Hierarchical Model

Kim et al. begin by considering N respondents from the dataset belonging to a group $g \in G$ [17]. Similarly to Kleiman-Weiner et al., Kim et al. state that this group could represent a country, culture or group within which customs and moral norms are shared. The moral principles of an individual i within group g are then drawn from a multivariate Gaussian distribution,

$$w_i \sim \mathcal{N}_D(w^g, \Sigma^g), \tag{4}$$

where w^g denotes the mean values of the group g over D dimensions and the diagonal of the covariance matrix Σ^g represents the difference between members of group g over the abstract principles. The covariance matrix Σ^g also holds information about the strength of relationships individuals place between abstract principles. For example, if an individual values infancy highly they may also value the elderly highly. The covariance matrix allows the learner to quickly infer moral principles of one dimension after inferring those of a highly correlated dimension [17].

Let $\boldsymbol{w} = \{w_1, ..., w_i, ...w_n\}$ be the set of moral principles of N respondents and the vector $\boldsymbol{\Theta} = \{\Theta_1^1, ..., \Theta_i^t, ..., \Theta_N^T\}$ represent the resultant states of i respondents over T scenarios. The decision made by respondent i is represented by a random variable Y_i^t . From this foregrounding, Kim et al. are then able to define the posterior distribution,

$$P(\boldsymbol{w}, w^g, \Sigma^g | \boldsymbol{\Theta}, \boldsymbol{Y}) \propto P(\boldsymbol{\Theta}, \boldsymbol{Y}) | \boldsymbol{w}) P((w) | w^g, \Sigma^g) P(w^g) P(\Sigma^g), \tag{5}$$

and likelihood,

$$P(\boldsymbol{\Theta}, \boldsymbol{Y}) | \boldsymbol{w}) = \prod_{i=1}^{N} \prod_{t=1}^{T} p_{ti}^{y_i^t} (1 - p_{ti})^{(1-y_i^t)}, \tag{6}$$

where p_{ti} is the probability respondent i chooses to swerve in scenario T, which is given by equation 2. Kim et al. define a prior over the covariance matrix Σ^g as a LKJ covariance matrix with parameter $\eta = 2$,

$$\Sigma^g \sim LKJ(\eta), \tag{7}$$

and specify a prior over the group weights,

$$w^g \sim \mathcal{N}_D(0, \Sigma^g). \tag{8}$$

Kim et al. continue to infer the model's individual weights, group norms and covariance matrices under the MAP estimate. We use Markov Chain Monte Carlo (MCMC) techniques to infer the posterior distributions over the model parameters. Making assumptions is essential when building data models, Bayesian methodology makes this explicit using priors. However, without conjugate priors inference becomes intractable in these models and we have to rely on approximative inference. In this paper we use MCMC methods which are correct in the limit. This paper looks closely at prior assumptions questioning if priors are chosen for computational simplicity or because they make sense.

VII. Preliminary Results

One advantage of using MCMC is its ability to approximate posterior distributions for model parameters. The posterior distributions for the model's group norm parameters over 5000 samples, with 1000 tuning samples, is shown in Figure 2. As these posterior distributions closely resemble normal distributions, MCMC techniques can be used to derive parameter point estimates. Kim et al. use point estimates to discuss the success of their model in inferring moral principles. Kim et al. compare the accuracy of their model in predicting individual responses against a benchmark model, Benchmark 1, which models respondent values along the abstract moral principles Λ. Kim et al. model the group weights,

$$w^f \sim Normal_D(\mu, \sigma^2 I), \tag{9}$$

ignoring hierarchical structure and assume that inferring the moral preferences of an individual does nothing to inform the inference of others. This work introduces Benchmark 2, that models individual weights,

$$w_i^l \sim [1, 1, 1, 1, 1, 1], \tag{10}$$

to represent the assumption that an individual assigns equal weight to each abstract principle.

Figure 3 shows the comparative predictive accuracies of our implemented model when compared to the two proposed benchmarks. We can see that the hierarchical model outperforms both benchmarks in predicting out of sample individual responses.

The comparative success of both Benchmark 1 and the Hierarchical model, when compared to Benchmark 2, seems to provide experimental evidence to support the existence of moral principles that are unique to individuals and govern the judgements we make in moral dilemmas. Interestingly, the variance between partitions is greatest within Benchmark 3, which suggests that whilst most humans operate in a utilitarian way there is great variability amongst individuals in their beliefs that all humans should be treated equally. This conclusion was also made by Faulhaber et al. upon conducting the original study [20]. The proposed hierarchical model therefore has offered a solution that deals with inter-person moral variability.

VIII. IMPORTANCE OF PRIOR ASSUMPTIONS

A. Non-informative priors

Machine learning is an instance of inductive reasoning, as such results that we have learned can never be proven right they can only be proven wrong. Together with the no free lunch theorem this means that the result of a machine learning method can only ever be interpreted relative to its assumptions. We question the assumptions made by Kim et al. and Kleiman-Weiner et al. when deriving their models. Prior assumptions have been identified as highly influential

Fig. 2. Inferred posterior distributions for group norm model parameters. Top Left: self preservation, Top Right: elderly, Middle Left: infancy, Middle Right: middle aged, Bottom Left: kneeling down, Bottom Right: pedestrian

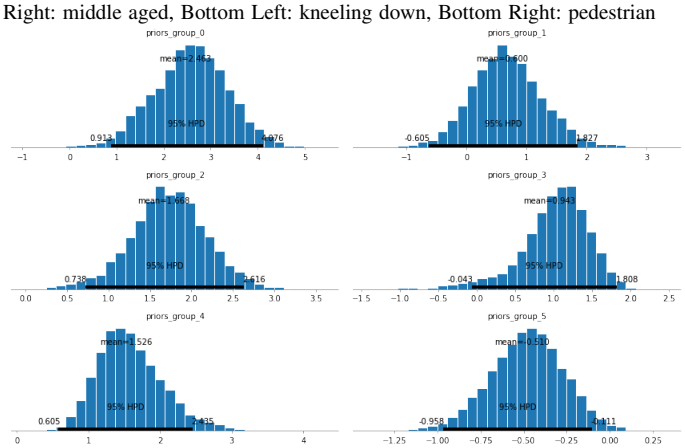
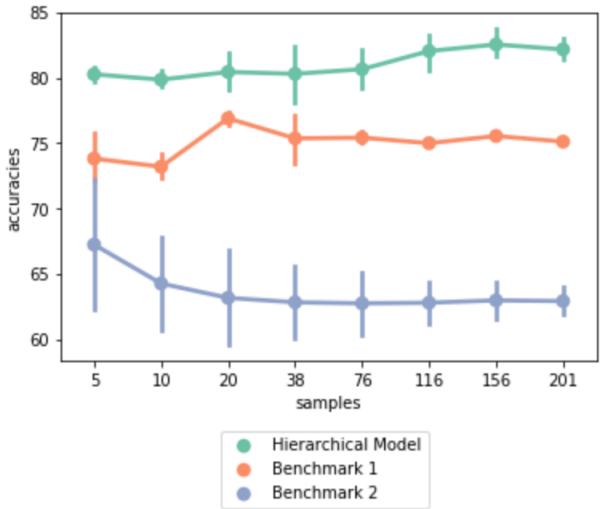


Fig. 3. Pointplot showing the differences in predictive accuracies between three different models. The Hierarchical model is the implemented model detailed above, Benchmark 1 shows results from the flat model, Benchmark 2 shows results from a model that assumes all individuals have equal weights over all moral principles. The Y axis accuracy shows predictive accuracy over individual decisions. The X axis represents the number of respondents in sample. The vertical bars show variance from cross validation.



on statistical models. Further, Kleimen-Weiner et al. identify the insufficient exploration of optimal prior distributions for their model and highlight this as an area for future work. We therefore make the hypothesis that the priors chosen by Kim et al. were chosen primarily for practicality and that, with further consideration, prior distributions could be found that more accurately reflect prior beliefs in moral theories. One of the biggest differences between Bayesian reasoning and traditional methods is the inclusion of subjective beliefs about a probability in calculations. Bayesian inference is therefore composed of both current and prior knowledge which act together to make up the posterior [26]. The prior distribution plays a central role in Bayesian inference, particularly in circumstances where the likelihood does not dominate the posterior, i.e. the volume of data is limited. Prior specification becomes particularly challenging when using hierarchical models for Bayesian inference, as these models require hyperparameters which require their own prior distributions [27]. We conjecture that as a society we still know little about human morality, where it originates, and how it varies from individual to individual or group to group [28]. We argue that the priors chosen in the model of Kim et al. make assumptions about the moral preferences of the underlying data that do not reflect theory. The decision of Kim et al. to model individual weights and group norms as precisely normally distributed is particularly questionable. Firstly, considering group norms: do we expect these values to be closely concentrated around a central value? How do we expect these values to be correlated? Secondly, considering individual weights: do we consider these values to be closely concentrated around the group norm with low mass in distribution tails? Is this an accurate reflection of society? We now present alternative prior distributions that attempt to address these issues.

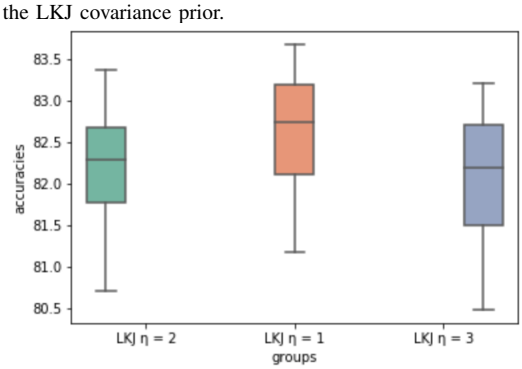
B. LKJ Covariance Prior

Kim et al. model the covariance of abstract moral weights as:

$$\Sigma^g \sim LKJ(\eta = 2) \tag{11}$$

However, there is little moral theory or evidence that accounts for the extent as to which moral principles are related to each other [29]. We propose experimentation with the LKJ distribution under different values to η to ascertain the prior over the covariance matrix that best suits the structure of the data. Figure 4 shows the results of our experimentation varying η. The boxplot shows that the model achieves greatest predictive accuracy when modeled using the LKJ covariance prior with $\eta = 1$. This implies our earlier hypothesis, that weakening prior assumptions more accurately reflects the structure of the data.

Fig. 4. Box plot showing results of experimentation with the parametrisation of the LKJ covariance prior.



C. The Half Cauchy Distribution

A further concern over Kim et al.'s approach is the normally distributed priors with narrow variance. The standard deviations for the mutivariate Gaussian distributions are currently normally distributed themselves as:

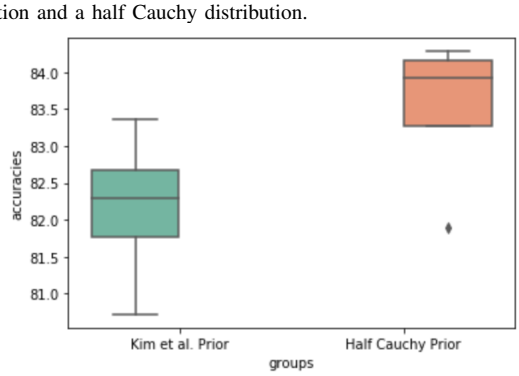
$$\sigma \sim \mathcal{N}(1, 0.1) \tag{12}$$

We propose the use of a weaker prior over the standard deviations of the model. This will challenge the idea that individual moral principles are tightly clustered around a group norm and the idea that group norms are constrained to lie between a narrow range. Gelman identified that many of the distributions that have been historically used for non-informative priors introduce levels of subjectivity into results [27]. Furthermore, Gelman proposes the half Cauchy density as a reference non-informative prior that should be used for the standard deviation term in a hierarchical model. The half Cauchy is a special case of the conditionally-conjugate folded-non central-t family of prior distributions for variance [30]. It has a wide peak at 0 and a scale parameter A. As A tends towards infinity, the half Cauchy distribution becomes a uniform prior on the standard deviation. Large and finite values of A represent a weak prior as the distribution has a gentle slope in the tail. We propose modeling the standard deviations of the group-norm parameters and individual-weights as,

$$\sigma \sim halfCauchy(10) \tag{13}$$

Figure 5 shows the comparative predictive accuracies between the standard deviation hyperprior specified by Kim et al. and the half Cauchy hyperprior. We can see that the half Cauchy prior statistically significantly improves the predictive accuracy of the model. This re-enforces our hypothesis that Kim et al.'s prior assumptions are too specific to represent the data. There is no concrete evidence to suggest that human moral beliefs are concentrated around a central group value. We therefore consider a new line of questioning into the use of normal distributions to model moral weights. We propose modeling individual weights with a general distribution that has increased mass in its tails.

Fig. 5. Box plot showing predictive accuracy of the hierarchical model, comparing modeling the standard deviation hyper prior as a narrow normal distribution and a half Cauchy distribution.



D. The T Distribution

In their work "Student-t Processes as Alternatives to Gaussian Processes" [31], Shah et al. highlight the popularity of the normal distribution due to its interpretability, large support base and the success of its empirical results. They continue to present Student-t processes as a family of elliptical processes that generalise the normal distribution, derive a definition for the Student-t-process by placing an inverse Wishart process on the kernel of a Normal distribution and give a definition of the Student-t-distribution,

$$(f(x_1), ..., f(x_n))^T \sim MVT_n(v, \phi, K), \tag{14}$$

where $K \in \Pi(n)$ is the covariance matrix and $\phi \in \mathbb{R}^n$ is the mean vector [31]. Shah et al. continue to stress the importance of the v parameter which controls how heavy tailed the distribution is. As v increases, the Student-t distribution converges to a normal distribution. We hypothesise that this property of the Student-t distribution may reflect a more accurate prior assumption of the group norms and individual weights. By using a distribution with more mass in the tails we are weakening the assumption on the model parameters, releasing them from the pressure to be concentrated around a central value. We therefore propose to model the group norms and individual weights as:

$$w^g \sim MVT(5, 0, \Sigma^g) \tag{15}$$

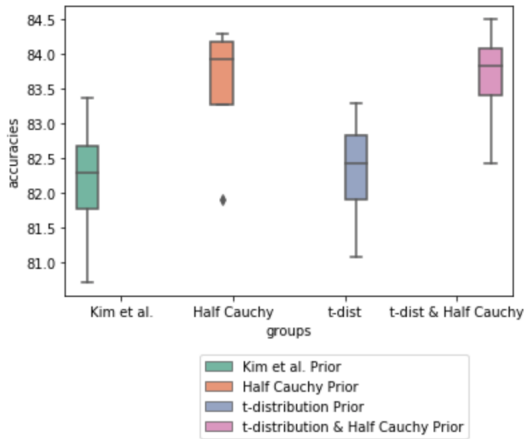
$$w^i \sim MVT(5, w^g, \Sigma^g) \tag{16}$$

Figure 6 is a box plot showing the predictive accuracy of the hierarchical model comparing modeling the individual weight parameters and standard deviation hyperprior as normally distributed; modeling the individual weight parameters as normally distributed and hyperpriors as half Cauchy; modeling the individual weight parameters with a Student-t distribution and hyperpriors with a normal distribution; modelling the individual weight parameters with a Student-t distribution and modeling the standard deviation hyperpriors with a half Cauchy distribution. Figure 6 shows that whilst the combined model achieves the greatest predictive accuracy over a single partition, the half Cauchy model remains the most robust with the lowest variance value across partitions. These results show the difficulty in selecting a prior distribution for a model. Is it more important to have a consistently performing model or achieve the greatest predictive accuracy? However, through demonstrating the sensitivity of the model to parameterisations, we have motivated the importance of carefully considering prior distributions, particularly in cases where prior knowledge is scarce. Furthermore, in the context of morality we must be careful to view the results in light of our dataset and stress the importance of further study in this area to generalise results.

IX. UTILIY FUNCTIONS

One of the central components of the model presented by Kim et al. is the use of utility calculus as a mechanism to compute welfare-trade offs that determine individual moral decisions. The idea of quantifying weights over moral principles that can be summed to determine an overall utility is synonymous with the ethical perspective of utilitarianism and places a big assumption over the ways in which humans

Fig. 6. Box plot showing predictive accuracy of Hierarchical model when modeled with different hyperprior distributions



make moral decisions. We continue to raise questions over the use of utility functions to infer individual moral principles and attempt to ground the experimental results found by our model in ethical theory.

A. Moral Principles

We question whether humans use moral principles to make decisions and indeed, whether these principles even exist. This debate is encapsulated by the ongoing Particularism vs. Generalism argument, which refers to the families of attitudes surrounding moral principles [32]. In his formative work, "Reasons as Defaults" [33]. J.Horty presents Generalism as the attitude that our ability to reach decisions is governed by general principles. Horty presents Particularism as the attitude that downplays the importance of general principles, instead emphasizing the role of receptivity to features of particular situations as the mechanism behind decsion making. In his work, Horty presents a counter-argument to the popular opinion given by J Dancy in support of Particularism [34]. Dancy argues that "a principle based approach to ethics is inconsistent with the holism of reasons" [34], where the "holism of reasons" refers to the idea that what counts as a reason for an action in a particular scenario may not apply to another scenario.

From the German Autonomous Vehicle dataset, we can clearly see evidence for the "holism of reasons". An individual who in one context chooses to save an elderly person over a child may do the opposite when given an alternate scenario. Contrastingly, Horty uses default logic to present a counter argument to Dancy [33]. Horty develops a framework that supports the "holism of reasons", whilst being built on a system of principles, which Horty terms defaults. Horty's central argument is based on the idea that reasons be thought of as being provided by defaults, where the default $A \rightarrow B$ means that A provides the reason for B. Horty continues to present the idea of priority relations between defaults, such that if default $\delta_1 : A \rightarrow B$ default $\delta_2 : C \rightarrow \not{B}$ and default δ_3 specifies the priority relation $T : \delta_1 \succ \delta_2$. This ordering means δ_2 has a higher priority than δ_1 and therefore, if there was a scenario where A and C both hold, δ_1 would be defeated [33]. Horty continues to expand this argument to derive a framework that uses the idea of defaults and priority relations to support the notion of "reason holism" [34]. We argue that our model implementation shows some experimental evidence for Horty's argument. From our model's inference of individual moral weightings we can define a priority default for each individual that ranks the defaults δ_1 : Child $\rightarrow$ Swerve; δ_2 : Elderly $\rightarrow$ Swerve; δ_3 : Self $\rightarrow$ Swerve; δ_4 : Kneeling $\rightarrow$ Swerve; δ_5 : Man $\rightarrow$ Swerve; δ_6 : Pedestrian $\rightarrow$ Swerve.

If an individual therefore has the priority default that gives δ_1 higher priority over δ_2, given a moral dilemma that presents a child and elderly man in each lane of the scenario, the priority relation would determine the outcome. Horty also presents the idea that priority relations are variable which supports the evidence we see in the data that decisions are not consistent over every scenario. Similarly, Guarini et al. present neural networks as an artificial intelligence tool that could, in the future, help to resolve the Particularism vs. Generalism debate [32]. We have seen how experimental evidence gathered by our model can be applied to the context of ethical theory. We do not pretend that this contribution has brought a resolution to the Particularism vs. Generalism argument, however, we have highlighted how exploration into artificial morality could contribute to historical ethical conundrums and help to illuminate ways forwards.

B. Do Humans Use Utility Functions to Make Decisions?

Utility theory has been used successfully in economics since its introduction by Von Neuman et al. in 1945 [35], who defined utility theory in terms of the attempts of the individual to obtain a maximum of utility or a maximum of profit. However, the differences between moral theory and behavioural decision theory is nuanced and difficult to define, thus questioning the use of utility theory in moral decision making. Further, this approach to moral decision making falls under the utilitarianism approach to ethics. Historically, researchers have identified utilitarianism and deontology as the two central competing modes of moral decision making [36]. Ditto and Liu discuss the differences between both perspectives and attempt to explain how they both contribute to moral decision making [37]. Ditto and Liu conjecture that humans tend to balance utilitarian outcomes with deontological principles [37]. For example, if an individual believes it is wrong to kill children and they are presented with a scenario that forces them to choose between saving five men or one child, this individual would increase their weighting of the child to be in line with their deontological standing. We incorporate the perspective of Ditto and Liu into the model proposed by Kim et al..

C. Incorporation of Deontological Statements Into Model

The previous section shows how weakening prior assumptions by generalising could lead to better results when prior beliefs in model parameters are limited. However, Gelman refers to these non informative prior distributions as "reference models" [27] with the view that these should be replaced with more informative prior distributions once more knowledge was available [27]. As part of the German Autonomous Vehicle study, Faulhaber et al. [20] ask participants a series of questions following their completion of the moral dilemma trials. These questions are intended to gauge the respondent's perspective on general ethical questions, i.e. "Protecting oneself should have priority over protecting others" and "It is sometimes acceptable to sacrifice elderly people in order to prevent the death of another person". Participants are asked to give each statement a score ranging from one to seven. A score of seven indicates that the participant strongly agrees with the statement whereas a score of one indicates complete disagreement. These questions obtain a sense of the individual's view of the rightness or wrongness of the action, removed from a specific context, which can be equated to their deontological perspective. We therefore propose to take the quantification of a participant's deontological beliefs and combine this into the model such that an individual's weights are modeled,

$$w_i \sim \mathcal{N}((w^g + \phi)/2, \Sigma^g), \tag{17}$$

using ϕ to represent the vector of individual i deontological beliefs.

Fig. 7. Box plot showing predictive accuracy of the hierarchical model with priors specified by Kim et al. [17] when compared to the model initialised with informative, deontological priors.

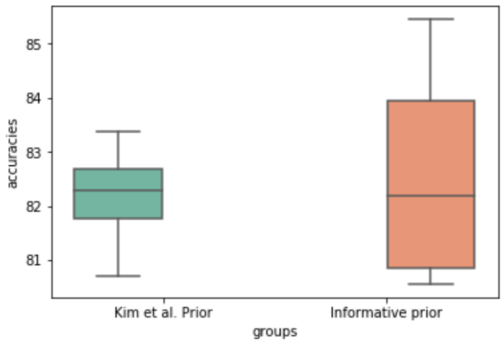


Figure 7 shows that by incorporating deontological statements into the model's prior distributions the model achieves its highest recorded predictive accuracy of 85.5% which supports our hypothesis that deontological statements are intrinsically linked to how we weight alternatives in moral dilemmas. However, the high variability across the partitions for this prior specification suggests that our hypothesis is not true over all individuals and that there is high variability amongst our data as to the extent in which individuals use deontological rules to shape their actions. If these two ethical perspectives are proven to be linked, this result could be invaluable in resolving the debate over optimum ethical theories.

X. CROSSS DOMAIN MODEL

One of our main motivations for implementing the model of Kim et al. was to test results on an alternate dataset. We have seen how the model proposed by Kim et al. has provably inferred moral principles over two alternate datasets. So far, however, we have only considered moral dilemmas in the context of trolley problems. In his PhD thesis, Awad highlights some of the limitations of using the trolley problem to frame moral theory [38]. Awad claims that trolley dilemmas are too simple, too rare, and unrealistic [38]. We do not, therefore, pretend to have implemented a model capable of extracting the full spectrum of human morality, however, we have used the trolley problem as a framing device to show the feasibility of this goal. We can therefore identify one critical area for future work as the applicability of the model of Kim et al. to other domains of moral theory [17]. We now present the application of Kim el.'s model to more general moral dilemmas.

XI. Transferring Model to a New Domain

The study conducted by Haykawa et al. [39] investigates the contribution of foreign language in responses to moral dilemmas. Hayakawa et al. present a wide range of scenarios to participants, which extend the idea of moral dilemmas beyond trolley problems. An example of these scenarios is shown in Figure 8. 20 variations of these scenarios were shown to respondents gathering their opinions on moral dilemmas ranging from coma patients to bear attacks. Every scenario poses a concluding statement and the respondent gives a binary response, which represents the extent to which they agree with the statement. From Figure 8 we can see that the presented moral dilemmas from Hayakawa et al. are more complex and realistic than the original dataset. Hayakawa et al. present six experiments, each containing a slight variation of dilemmas, and a sample size of 224 in a laboratory environment. We propose to apply the model of Kim et al. to the data generated by one of the experiments of Hayakawa et al.

One of the biggest challenges in transferring the model of Kim et al. to the new domain is redesigning the mapping of character vectors to the abstract feature space. Figure 8 shows the complexity of the moral dilemmas as presented by Hayawaka et al. However we believe that, similarly to trolley problems, these moral dilemmas can be entirely described by a set of abstract features. The example from Figure 8 could, for example, be entirely described by the abstract features, criminal and torture, for one decision and property damage for the other. This, we believe, encapsulates the spirit of the moral dilemma. Due to the high variability between the 20 moral dilemmas presented in the experiments of Hayakawa et al. the resulting abstract feature vectors are much more difficult to design. We

therefore experiment with different abstract feature vectors, encapsulating different numbers of moral feature parameters that can be used to describe individual moral preferences of the dataset.

After defining the abstract feature vectors, we use the model parametrisation of Kim et al., We are then able to infer individual moral preferences over the 224 respondents and use these to predict out of sample decisions. Figure 9 shows the predictive accuracy of our implemented model using abstract feature vectors of varying length to represent the dataset from Hayakawa et al. Figure 9 shows how the predictive accuracy of the model increases with the number of abstract moral features used to describe underlying moral dilemmas, where error bars represent the variance in accuracy over different permutations of chosen abstract features. Our results show that the accuracy decreases as the number of parameters exceeds seven. Furthermore, when we attempt to define eleven abstract moral features the model encounters sampling issues and does not converge. This behaviour can be explained by the incapability of the model in dealing with sparse abstract feature vectors. Furthermore, we cannot conclude that seven abstract moral features are sufficient to describe individual moral preferences. In fact, these results show the impact of placing different assumptions over how many, or which, abstract features are used to describe moral dilemmas and represents a significant challenge to the method of Kim et al. if the model is to be expanded to learn moral preferences over more complex moral dilemmas.

Our results have shown how significant model selection is in statistical analysis. A commmon approach of data scientists when tackling this problem is to fit multiple models to their data and then select one depending on model comparison heuristics [40]. Gershman et al. introduce Bayesian Non-parametric models (BNP) that offer an alternative approach to this problem [40], where non-parametric models allow model complexity to grow as more data is observed. Gershman et al. continue to discuss latent feature models, which describe observations composed of a number of hidden factors. The model as proposed by Kim et al. is an example of a finite latent fixture model, whereby the number of abstract moral features is specified prior to inference. However, our results have shown that it is not always straightforward to select the number of latent factors that perfectly describe data. Griffiths et al. [41] introduce a BNP model, the Indian Buffet Process (IBP), that allows for an unbounded number of latent features.

We have found that a serious limitation in the model proposed by Kim et al. is the ambiguity of selecting the number of latent features that are capable of completely describing the moral data. We have suggested non parametric methods such as IBP could offer a solution. However, we must return now to the "No Free Lunch Problem" and remind the reader that there is no optimal solution to this problem. One significant disadvantage of non-parametric models is their intractability. Furthermore, future work in this area must work out how to balance representation of complex data with computational power, a problem that has plagued data science since its origin. We therefore conclude with the argument that the future success of engineering ethical machines is intrinsically dependent on the resolution of classical mathematical problems.

XII. AGGREGATING RESULTS

Finally we turn our attention to ethical machines. Can models of human moral decision making be used to automate ethical decisions? Kim et al. identify that an expansion of their model could be to investigate the design of an ethical agent that, when faced with a specific moral dilemma, is capable of making a decision that maximises the overall social utility of a population. Noothigatttu et al. [22] use recent developments in computational social choice to suggest a way of aggregating individual preferences to identify an overall desirable choice for a given ethical dilemma.

A. Ethics without Rules

Noothigattu et al. identify the central challenge that has so far prevented progress in automating ethical decision making as the lack of ground-truth principles. In most successful AI applications, there exists a bonified right answer that is provable from data. However, in the world of ethics, there is no correct definition of ethical behaviour. Dwork et al. [42] claim that in the absence of ground truth rules, the optimum decision should be agreed upon by society. Kim et al. expand this idea to suggest aggregation of individual preferences to automate ethical decision making. Further, when faced with the dilemma of best incorporating human behaviour into an automated ethical system, Greene et al. consider a solution in the longstanding field of preference aggregation [43]. Connitzer at al. extend this idea by suggesting that we let models of human moral decision making vote over the best possible outcome [44]. Noothigattu et al. use these studies to propose a framework of ethical decision making, utilising computational social choice [22].

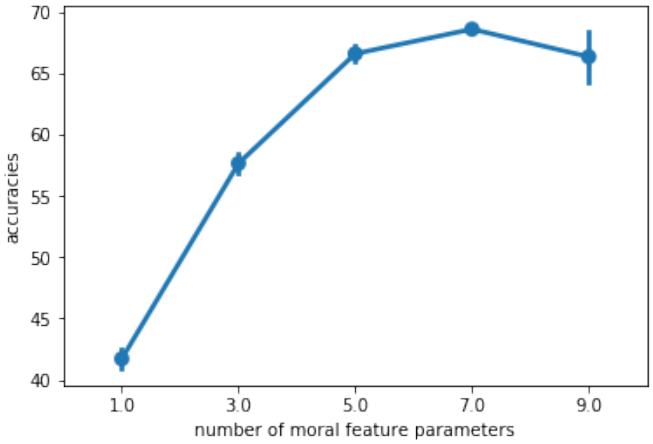
B. Voting Based Ethical Decision Making

Noothigattu et al. summarise their framework into three key stages: 1. Learning 2. Summarisation 3. Aggregation. The first stage learns individual models of preferences using pairwise comparisons over all possible scenarios. The second stage combines all individual models into a combined model of preferences. The third stage uses the summarised model such that, at run time, when presented with an ethical dilemma, a voting rule is applied to aggregate preferences into a collective decision [22]. One of the biggest challenges faced by Noothigattu et al. is the selection of a voting process that aggregates individual preferences into one overall winner. Traditional methods consist of techniques such as the Borda count, which determines an outcome by giving each alternative, for each individual, a score that is then summed over all individuals such that the

Fig. 8. An example moral dilemmma posed to respondents in the Hayakawa et al. study [39]

> You are a police officer, and have recently caught a criminal you have been hunting for some time. He has made many bombs: some that have already exploded and some that have not exploded yet.
>
> He places bombs outside cafes and they explode when no one is around. His bombs are inside paint cans so that they spray nearby objects with paint. He has sprayed many cafes with paint and might spray many more.
>
> Now that the criminal is caught, you want to know where the unexploded bombs are so you can stop them. The only way to make him talk is to use torture. For example, holding his head under water and beating him.
>
> Is it morally correct for you to use torture in order to prevent further property damage, even though the criminal will suffer?

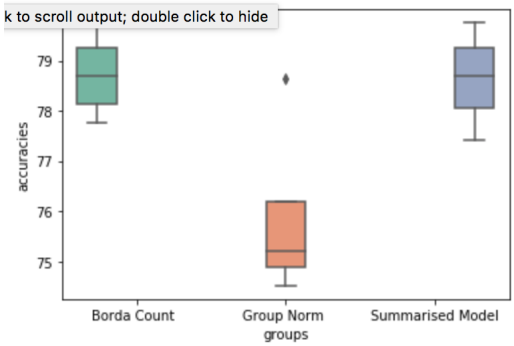
Fig. 9. Pointplot showing the relationship between model predictive accuracy and the number of abstract moral features used in parametrisation on the Hayakawa et al. dataset [39]



alternative with the highest score wins. Noothingattu et al. note the potential computational cost of voting techniques such as the Borda count when used on large datasets [22]. They therefore propose a more computationally efficient summarised model [22].

We propose the use of Noothigattu et al.'s summarised model to question of the ethicality of Kim et al.'s model of moral behaviour. Can ethical behaviour actually be defined as the average behaviour of humans in moral dilemmas? Furthermore, how representative is this framework? How many individuals would agree with the decisions made by the aggregate model? We evaluate this question by using Noothigattu et al.'s proposed aggregation of the summarised model to predict individual out of sample decisions. Given an out of sample scenario, Noothigattu et al.'s odel is applied to aggregate all individual sample decisions into a final majority decision. This decision is then compared to that of the out of sample respondent. Figure 10 shows how the Borda count is the most representative aggregation process but is nearly matched by Noothigattu et al.'s summarised model. Furthermore, the aggregated model matches the decisions of 78% of the population. This overwhelming majority suggests that this is a good starting point to begin engineering ethical machines that behave most desirably according to the majority.

Fig. 10. Boxplot comparing the out of sample predictive accuracy of model under different aggregation methods when using different methods of aggregation to generate universal moral weightings.



XIII. DISCUSSION

This work has replicated the exciting results produced by state of the art in modeling ethical decision making. We have extended these studies by combining philosophical and mathematical theory to suggest alternative model parameterisations, that both reflect ethical theory more closely and improve quantiative results. However, above all we have raised a multitude of potential barriers to this approach to automating ethical decision making. Further, this work has established the enormity of the ethical AI challenge and has throughout emphasized the importance of questioning quantitative results. One key challenge this work has uncovered is the challenges of interpreting machine learning results based on data that described tacit human knowledge. Results can only be interpreted relative to their associated assumptions and making assumptions of human ethics and morality is very challenging. The richness of this work therefore lies in its cautionary rhetoric and how we apply the presented analytical techniques to future work. We therefore conclude by presenting some possible directions for future work.

A. *Oversimplification of decision making*

One significant criticism of the implemented model is the decision to model decision making as a linear combination of moral preferences that remains invariant over time. However, we question whether a richer model of moral decision making would be achieved by modeling, not just the decision outcome, but also, the cognitive process of deliberation. Busemeyer et al. identify the shortcomings of historical methods of modeling decision making using expected utility theory [45]. They draw on the psychology theory of James [46] to highlight the importance of

cognitive deliberation processes in making decisions under uncertainty. Busemeyer et al. build on historic work of static models of decision making to propose a dynamic model, using a drift diffusion model to show how preference relations change as a function of deliberation time.

An alternative oversimplification of decision making assumed by the Kim et al. model is the decision to model decision making processes as utility functions representing underlying moral preferences. Furthermore, there has been huge debate in the ethics sphere over "Principilism", the practice of using principles to replace moral theories. Clouser et al. argue that misguided moral principles obscure moral reasoning by misrepresenting and over simplifying moral theories [29]. We can see this oversimplification of moral theories within our implementation. When defining their utility calculus, Kleiman-Weiner et al. include both a linear representation of moral preferences **and** another term representing other abstract individual qualities that affect moral decision making such as empathy. Furthermore, when defining their model, Kim et al. ignore this additional term and focus on inferring weights over moral principles. An extension of this model would therefore be incorporating a method of modeling the additional source of moral valuation.

Considering empathy as a factor that shapes moral decision making in the trolley problem has been approached in the literature by Wilson et al. [47] who begin by modeling decision making as utility scores, a linear combination of propositions that have an associated weight. To account for empathy in their model, Wilson et al. introduce a new weight on the utility of a proposition that determines the empathetic response of an action. This empathy model could be used to enrich the model of Kim et al. by accounting for individuals whose behaviour could not be explained by moral principles alone: a naturally empathetic person, a mother who had recently given birth or a husband who had recently had an argument with his wife.

B. Towards a Hybrid Model

This work discussed the difference between bottom-up and top-down approaches to engineering ethical machines. We have presented a bottom-up approach to learning human morality and a method for combining individual models into an aggregated model of morality. The next stage in engineering an ethical machine would therefore be to incorporate ethical or legislative rules into the system presented in this work to produce a hybrid ethical model. Noothigattu et al. recommend that ethical rules could control behaviour of the machine. However, they claim it impossible to specify ethical rulings that cover every possible scenario, in which case the aggregated model of human moral principles would control behaviour [22]. Loreggia et al. also propose an alternative method for encoding ethics within models of human moral behaviour [48]. Their proposed system uses CP-nets to evaluate whether human preferences are compatible with specified ethical principles. CP-nets were presented by Boutiller et al. in 2004 as a graphical representation of preference data and allow for the qualitative instead of quantitative assessment of preferences using preference relations [49]. Loreggia et al. continue to construct two alternative CP-nets, with the first over human moral preferences and the second over ethical principles. This formulation allows Loreggia et al. to calculate the distance between both CP-nets and a way of measuring whether moral principles are "close enough" to ethical rules. This would be an interesting extension of this work. However, the question remains, what set of ethical rules should be used? We propose the use of the recently defined German ethical guidelines for autonomous vehicles commissioned by the German federal government and the first of its kind to standardise the behaviour of autonomous vehicles in ethical dilemmas [50]. The incorporation of these guidelines as ethical rules would show the difference between human behaviour and legal requirements defined by lawyers which could make for a very interesting comparison.

C. Quality of data

An argument that uses inductive reasoning can never be proven right or wrong but can be strengthened through use of quality evidence, this means that we need lots of representative data to justify our method. We continue to re-iterate that the results and arguments presented in this work are contextualised to our datasets and are by no means a resolution to the challenge of designing ethical machines. Furthermore, future work must continue to gather quality data. One specific issue surrounding the quality of data used by this study is lack of fully representative data. Both the Moral Machine dataset [19] and German Autonomous Vehicle dataset [20] represent the moral beliefs of Western cultures. If future studies wish to use models of morality to automate ethical decision-making, consideration must be made to ethnic and socio-economic groups that are traditionally under represented. Importantly through this work we have highlighted the challenges of generating enough quality data surrounding human ethics which must be addressed if future inductive models are to be trusted.

D. Ethicality of voting rules

We wish to draw attention to the ethicality of utilising a voting rule to aggregate individual moral decisions. Whilst we have seen that the aggregated model is representative of the majority of our data population, there are other ethical arguments that should be considered. Two of these central arguments are presented in the work of Risse, in his informative work "Arrows Theorem, Indeterminacy, and Multiplicity Reconsidered" [51]. Risse presents the indeterminate problem as a historical negative result in social choice theory. The theorem highlights inconsistencies in preference aggregations such that a sequence of pairwise votes may not lead to a majority vote but to a cycle. Consider, for example, in the context of our moral data, a population who prefers the elderly to infancy and infancy to self preservation but prefers self preservation to elderly. Risse argues that if cycles occur, voting rules do not produce a result and leaves the

decision indeterminate [51]. The second ethical challenge of voting rules presented by Risse is that of the multiplicity method, which refers to the problems incurred due to the existence of a wide variety of mutually inconsistent voting rules. Upon the selection of a particular voting rule there are guaranteed "winners" and "losers" whose identity is dependent on the specific voting rule [51]. Future work that attempts to aggregate moral preferences must therefore consider the underlying properties of their selected voting rule. An interesting extension of this work would therefore be to compare the decisions generated by the summarised model [22] to other traditional voting rules.

E. Algorithmic Fairness

Currently, one controversial area within ethical artificial intelligence is the concept of algorithmic fairness. A 2016 study found that the algorithm used in criminal sentencing, to assess risk of recidivism, was displaying signs of racial bias [52]. The fact that artificial intelligence has already spread to areas in which it has a big impact on human lives means that developers must start thinking beyond the predictive accuracy of their systems and consider the social implications [53]. Up to this point, we have considered aggregating models of individual moral preferences to automate decision making. We have considered representation as an ethical issue. However, we now present the ethical issue of fairness. Learning a model of human morality is inherently biased. We have seen how our data population are likely to place higher preference on humans with certain characteristics. Is it therefore ethical to use models of human morality in automating ethical decision making? Many critics would argue no, that humans are not an example of ethicality and that ethical intelligent systems should be designed without individual moral influence. Advocates, however, would argue that ethics is a human construct and therefore can only be considered within human conduct. If we are to use human moral behaviour to automate decision making, extensions to models must be made to prevent the use of protected attributed in decision making [53]. In their paper, "Counterfactual Fairness", Kusner et al. describe protected attributes as variables that must not be discriminated against, relative to a particular system. Kusner et al. present counterfactual fairness as the idea that a decision is only fair if it is fair in both the real world and a counterfactual world, where the target individual belongs to an alternative demographic group. An interesting extension of Chapter 11 would incorporate this idea into the summarisation process, such that only individual preferences that were counterfactually fair could be included in the summarisation process.

The results of the implemented model show that, within our data population, individuals place a higher weight on the infancy moral principle when compared to the elderly moral principle. This result shows that individuals are likely to discriminate based on potentially protected attributes, such as age. The ability of the model to infer quantified rankings over moral principles could, potentially, be expanded to recognize certain biases in certain populations, or software. An interesting extension of this project would be to apply the model of Kim et al. to the COMPAS open source dataset used for the recidivism study cite [52]. By assigning a weighting over race moral principles, we hypothesise that the model would be able to extract racial bias. A model that was capable of verifying the neutrality of a piece of software or a person would be invaluable in industries such as law, banking and recruitment.

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